IB MATH AA SL

Lesson 28

Probability & Venn Diagrams

Statistics & Probability · 60-minute lesson

Syllabus SL 4.5–4.6 · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. P(heads) on a fair coin?
2. P(rolling a 6) on a fair die?
3. P(not a 6) on a fair die?

Answers (click to reveal)

1. 1/2
2. 1/6
3. 5/6

What we're learning today

LEARNING OBJECTIVES

1. **Calculate** theoretical probability.
2. **Use** Venn diagrams for events.
3. **Apply** the addition rule for $P(A \cup B)$.
4. **Use** the complement $P(A') = 1 - P(A)$.

SUCCESS CRITERIA — I CAN...

- ✓ work out P(event) = favourable / total
- ✓ place data in a Venn diagram
- ✓ use $P(A \cup B) = P(A)+P(B)-P(A \cap B)$
- ✓ use the complement rule

Basic probability

DEFINITIONS

$$P(A)=\frac{n(A)}{n(U)}$$

$$P(A')=1-P(A)$$

n(A)
number of favourable outcomes

n(U)
total number of outcomes

Complement
'not A' = 1 – P(A)

SCALE

0
impossible

1
certain

All P
add to 1 over the whole sample space.

Simple probability

QUESTION

A bag has 4 red and 6 blue balls.

Find P(red) and P(not red).

SOLUTION

Total = 10. P(red) = favourable/total:

$$P(red) = \frac{4}{10} = \frac{2}{5}$$

Complement: P(not red) = 1 − P(red):

$$P(not\ \ red) = 1 - \frac{2}{5} = \frac{3}{5}$$

Your turn — probability

ANSWER IN YOUR BOOK

1. A die: P(even)?
2. A bag of 3 green, 7 yellow: P(green)?
3. If P(A) = 0.3, find P(A′).

Answers (click to reveal)

1. 1/2
2. 3/10
3. 0.7

Venn diagrams & the union

ADDITION RULE

$$P(A \quad B)=P(A)+P(B)-P(A \quad B)$$

∩

intersection — 'both A and B'

∪

union — 'A or B (or both)'

Subtract

the overlap, so it isn't counted twice

MUTUALLY EXCLUSIVE

No overlap: $P(A\cap B) = 0$

Then

$P(A\cup B) = P(A) + P(B)$

Use the addition rule

QUESTION

P(A) = 0.5, P(B) = 0.4, P(A∩B) = 0.2.

Find P(A∪B).

SOLUTION

Addition rule:

$$P(A \quad B)=0.5+0.4-0.2$$

Evaluate:

$$P(A \quad B)=0.7$$

EXAM-STYLE QUESTION

Exam-style question

In a class of 30, 18 study French, 12 study Spanish, 5 study both.

(a) How many study neither? [3]

(b) Find P(French only). [2]

MARK SCHEME — reveal after students attempt (tutor only)

(a) French only 13, Spanish only 7, both 5 → 25 study a language (M1)(A1) → neither = 5 (A1)

(b) P(French only) = 13/30 (M1)(A1)

Common mistakes

- **Double counting**
  Subtract $P(A \cap B)$ in the union rule.

- **$\cap$ vs $\cup$**
  $\cap$ = 'and' (overlap); $\cup$ = 'or'.

- **Probabilities**
  must lie between 0 and 1.

- **Complement**
  $P(A') = 1 - P(A)$, not $-P(A)$.

Mixed practice

QUESTIONS

1. Die: P(less than 3)?
2. $P(A)=0.6$, $P(B)=0.3$, $P(A\cap B)=0.1$. $P(A\cup B)$?
3. If P(rain)=0.25, P(no rain)?
4. Mutually exclusive A,B: $P(A)=0.4$,$P(B)=0.5$. $P(A\cup B)$?

Answers (click to reveal)

1. 2/6 = 1/3
2. 0.8
3. 0.75
4. 0.9

Plenary & exit ticket

KEY FORMULAS

$$P(A) = \frac{n(A)}{n(U)}$$

$$P(A \quad B) = P(A) + P(B) - P(A \quad B)$$

EXIT TICKET — before you leave

1. Die: P(a 5)?
2. P(A)=0.7 → P(A′)?
3. ∩ means 'and' or 'or'?

Answers (click to reveal)

1. 1/6
2. 0.3
3. and

Self-assess:

- need more help
- nearly there
- confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:
Probability & Venn diagrams.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 29

Conditional probability & tree diagrams.

$$P(A \mid B) = \frac{P(A \quad B)}{P(B)}$$